Progress Check

Write down the slope of each of the lines shown on the grid below. One is solved for you.

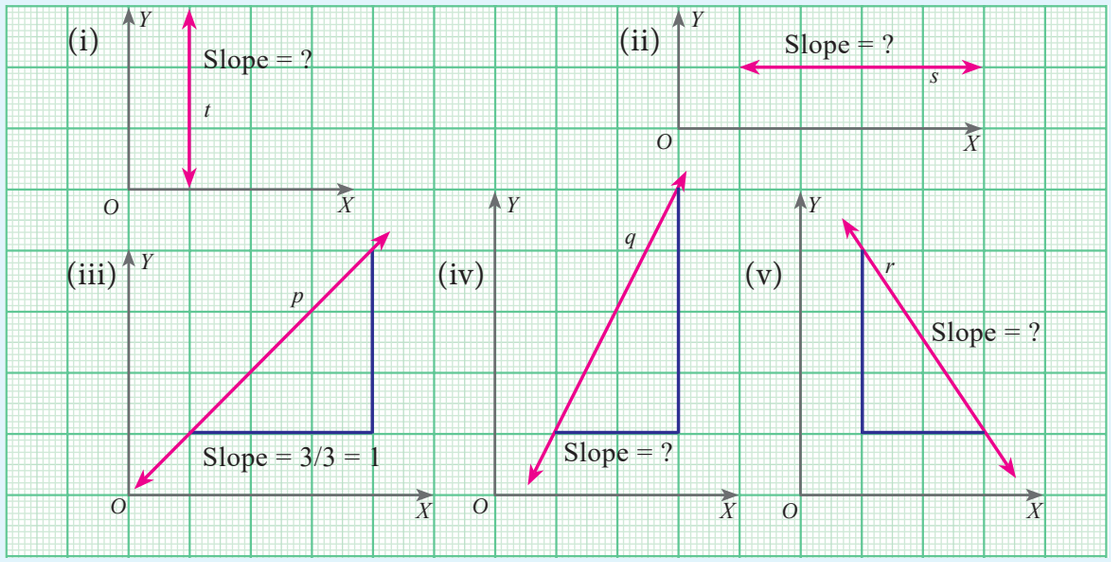


Fig. 5.20

Solution (iii) Slope of the line $p = \dfrac{\text{Difference in } y \text{ coordinate}}{\text{Difference in } x \text{ coordinate}} = \dfrac{3}{3} = 1$

5.4.2 Slopes of parallel lines

Two non-vertical lines are parallel if and only if their slopes are equal.

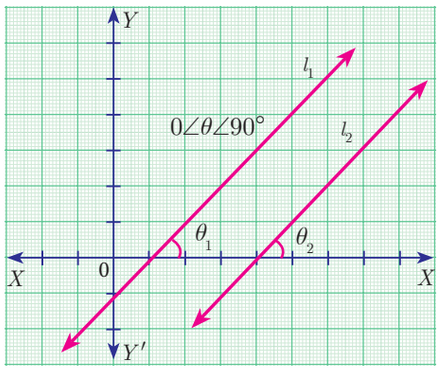


Fig. 5.21

Let l_1 and l_2 be two non-vertical lines with slopes m_1 and m_2 respectively.

Let the inclination of the lines with positive direction of X axis be θ_1 and θ_2 respectively.

Assume, l_1 and l_2 are parallel

$$\theta_1 = \theta_2 \text{ (Since, } \theta_1, \theta_2 \text{ are corresponding angles)}$$

$$\tan\theta_1 = \tan\theta_2$$

$$m_1 = m_2$$

Hence, the slopes are equal.

Therefore, non-vertical parallel lines have equal slopes.

Conversely

Let the slopes be equal, then $m_1 = m_2$

$$\tan\theta_1 = \tan\theta_2$$

$$\theta_1 = \theta_2 \text{ (since } 0 \le \theta_1 \le 180°,\ 0 \le \theta_2 \le 180°)$$

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